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B.Tech. Degree IV Semester Examination April 2019

CS 1406 DATA COMMUNICATIONS (2012 Scheme)

Time: 3 Hours

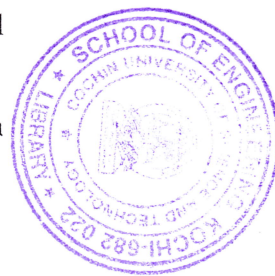
Maximum Marks: 100

PART A

(Answer **ALL** questions)

(8 × 5 = 40)

- I. (a) List out the advantages of digital transmission over analog transmission.
- (b) Define channel capacity. A data is to be transmitted over PSTN using 8 levels per signal elements. If the bandwidth is 3000 HZ, deduce maximum data transfer rate.
- (c) Illustrate PCM technique with the help of a neat diagram.
- (d) Exemplify any two block coding schemes used for digital signal transmission.
- (e) Summarize the three different transmission modes possible in HDLC.
- (f) Analyse the significance of Hamming Code in Forward Error correction using a suitable example.
- (g) Write short note on CDMA.
- (h) Discuss the major components of a telephone network.



PART B

(4 × 15 = 60)

- II. (a) Summarize the various transmission impairments. (10)
- (b) Explain wireless transmission. (5)
- OR**
- III. (a) Describe the components in communication model with a neat diagram. (5)
- (b) Explain any three different types of guided transmission media. (10)
- IV. (a) Define line coding. Encode the data 110110 using NRZ-L, NRZ-I, Manchester, Differential Manchester and AMI. (10)
- (b) Differentiate AM and FM. (5)
- OR**
- V. (a) Construct a Huffman code tree with probabilities given as 0.4, 0.19, 0.16, 0.15, 0.1. Also find the code for each symbol and calculate its average code size and variance. (9)
- (b) Discuss any two variants of PSK technique. (6)
- VI. (a) A series of 8 bit message 11100110 is transmitted across a data link using a CRC for error detection. A generator polynomial of 11001 is to be used. Illustrate the following: (10)
 - (i) CRC Generation Process
 - (ii) CRC Checking Process
- (b) Explain the concept of redundancy in error detection. (5)
- OR**
- VII. Summarize the three categories of ARQ error control mechanism. (15)
- VIII. (a) Explain the role of CM and CMTS in cable networks. (5)
- (b) With the help of necessary figures explain transmitter and receiver system in frequency hopping spread spectrum. (10)
- OR**
- IX. (a) Compare frequency division multiplexing and time division multiplexing. (5)
- (b) Explain FDM with the help of necessary sketches. (10)